Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21311

F-411

B.C.A. (Third Semester) Examination, 2023-24

(New Course)

Paper - BCA-301-N

COMPUTER ORIENTED NUMERICAL ANALYSIS

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Find a root of the equation $x^3-4x-9 = 0$, using bisection method correct to three decimal places.

   (b) Broadly, explain the use of Newton's divided difference interpolation formula.

2. (a) Find a real root of the equation $x^3-x-1=0$ using Newton Raphson method, correct to four decimal places.

   (b) What are truncation errors in interpolation method?

3. (a) Solve $x^3-9x+1= 0$ for the root lying between 2 and 4 by the method of false position.

   (b) Define Simpson's 1/3rd rule.

4. (a) Find solution using Newton's forward difference formula

x :	1891	1901	1911	1921	1931
f(x)	46	66	81	93	101

X= 1895

(b) Why Gauss Jordon method requires double the elementary row operations than Gauss Elimination method?

5. (a) Give comparison of Runge-Kutta and predictor corrector method.

(b) Evaluate $\sqrt{12}$ by Newton's formula.

6. (a) What is Nonlinear Regression? Comparison to Linear Regression with example.

(b) Explain Gauss elimination method with example.

7. (a) Evaluate $I = \int_0^1 \frac{1}{1+x} dx$ correct to three decimal places using trapezoidal rule.

(b) Using Euler's method, find an approximate value of y corresponding to x=1, given that $dy/dx = x + y$ and $y = 1$ when $x = 0$.

8. (a) Using Runge-Kutta method of order 4, find y for $x = 0.1$ 0.2, 0.3 given that $dy/dx = xy + y^2$, $y(0) = 1$. Find the solution at $x = 0.4$ using Milne's method.